

# # DBMS

## # ATTRIBUTES — column of Table

① simple → not divide more (Ex Age)

② Composite attribute → can be Divide  
(Ex name)  
first name last name

③ MULTI ~~VAL~~ VALUE →

④ DERIVED → derived from other attribute

Ex → DOB

↓  
Age

⑤ Key → unique attribute

# Key

① Super Key  $\rightarrow$  combination of attribute so that it can uniquely identify.

| Roll NO | DOB        | Age | Name | MOB        |
|---------|------------|-----|------|------------|
| 1       | 12/11/2005 | 21  | A    | 9876543210 |
| 2       | 6/4/2005   | 20  | B    | 1234567890 |

EX  $\rightarrow$  Roll NO + MOB

② candidate Key  $\rightarrow$  all are super Key but it can have minimal column so that it uniquely identify

③ Primary Key  $\rightarrow$  single column & to uniquely identify

④ foreign Key  $\rightarrow$

⑤ ALTERNATE Key  $\rightarrow$  Primary Key other than left

# # ACID PROPERTIES

o set of properties for reliable Transactions

① Atomicity —? Transaction: either fully happens or not at all.

② consistency — Database always stays in a valid state.

③ isolation — multiple Transactions don't affect each other.

④ Durability — Data same permanent  
→ even sys crash.

# Transaction log → detail Journal that records every change made by Transaction.  
Ex use for recovery, redo,

# DATA INTEGRITY: FAZHAMAM

It means Data overall accuracy, completeness and reliability of data.

Domain integrity → restrict the format Type and volume of data.

ENTITY integrity → Data not repeated multiple time.

REFERENCAL INTEGRITY → remove duplicates records

user defined integrity → fulfill the specific requirement

1) NOT NULL → field cannot be empty

2) UNIQUE → field cannot have same value

3) PRIMARY KEY → field cannot be empty and unique

# # NORMALISATION

Technique To reduce Redundancy in a table

↓  
Duplication

- ① Anomalous -  
( ) Insert  
Update  
Delete

Types

- ① 1NF  $\rightarrow$  NO multivalued attribute
- ② 2NF  $\rightarrow$  1NF  $\rightarrow$  NO partial dependency
- ③ 3NF  $\rightarrow$  2NF  $\rightarrow$  NO Transitive dependency
- ④ BCNF  $\rightarrow$  3NF  $\rightarrow$  LHS of functional dependency must be part of super key

|           |
|-----------|
| SUB       |
| JAVA      |
| C         |
| C++, JAVA |

multivalued attribute

| ROLL ID | NAME | SUB- ID | MARKS |
|---------|------|---------|-------|
| 1       | A    | 1       | 78    |
| 2       | B    | 2       | 80    |
| 3       | C    | 1       | 89    |

Non prime attribute can be determined by part of primary key  $\rightarrow$  Partial dependency

Transitive dependency  $\rightarrow$  Non prime attribute depend on other non prime attribute

~~Index~~  $\rightarrow$

INDEX  $\rightarrow$  data structure that helps up retrieval info

# clustered index  $\rightarrow$  sort entire ~~by~~ Table  
by one key

# Non clustered  $\rightarrow$  quick sort list  
look up

| NAME | Age | Sex | list |
|------|-----|-----|------|
| 85   | 1   | 2   | 1    |
| 08   | 2   | 8   | 2    |
| 18   | 1   | 3   | 3    |

get similar at one table only  
get similar at one table only

data will not be merged  
into one table  
data

~~Table~~

get table with index value  $\rightarrow$  Table